Week 14

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Mobile:

Data Structure Workouts
1. Learn about Bubble Sort, Insertion Sort, Selection Sort, Quick sort and Merge sort. Complete at least three sample workouts in each of them. 2. Learn the concept of Stack and Queue. Complete at least three sample workouts in each of them & do at least 3 problems from any competitive coding websites (Hacker Rank, Code Chef, Leet code, Algo Expert, etc.) a. PUSH, POP and Display elements in Stack b. Enqueue, Dequeue and Display elements in Queue 3. Learn the concepts of Hash Table. Complete at least 3 sample workouts. 4. Learn about the Applications of all structures you covered this week
Explored five sorting algorithms—Bubble, Insertion, Selection, Quick, and Merge Sort—and implemented three sample workouts for each. These algorithms, ranging from basic to advanced, were applied to sort arrays of integers, strings, dates, characters, and decimals. The task enhanced my proficiency in understanding and utilising diverse sorting methods efficiently. *Link to the folder containing code and screenshot of the output*
I successfully grasped the principles of Stacks and Queues, essential data structures. For Stacks, I implemented PUSH, POP, and displayed elements, adhering to the Last In, First Out (LIFO) approach. In Queues, I executed Enqueue, Dequeue, and displayed elements, following the First In, First Out (FIFO) principle. Additionally, I tackled three problems on competitive coding platforms, applying these concepts to real-world problem-solving scenarios. This task not only enhanced my understanding of Stacks and Queues but also showcased their practical utility in coding challenges. *QUEUE* *STACK*

Data Structure Workouts
I have successfully learned the concepts of Hash Tables, a fundamental data structure. Hash Tables employ a hash function to map keys to specific indexes, facilitating efficient data retrieval. Three sample workouts were completed, focusing on implementing key operations like insertion, retrieval, and deletion within the context of Hash Tables. This task deepened my knowledge of Hash Tables and their applications in organizing and accessing data with optimal time complexity. *Link to the folder containing code and screenshot of the output*
This week's exploration of fundamental data structures, including Stacks, Queues, and Hash Tables, provided a versatile toolkit for various applications. Stacks and Queues are foundational in managing data flow, with Stacks proving beneficial in functions like expression evaluation and Queues being pivotal in tasks such as process scheduling. Hash Tables, on the other hand, find widespread use in efficient data retrieval and storage, offering a key-value mapping that suits applications like indexing, caching, and symbol tables. The acquired knowledge of these structures equips me to address diverse computational challenges across different domains.